



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Commercial Satellite and LEO Constellation Cyber Threat Landscape

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

Table of Contents

1. Executive Summary
2. Intelligence Requirements
3. Source Review & Confidence
4. Threat Activity Overview
5. Affected Sectors and Exposure
6. Aviation / Aerospace Sector Impact Analysis
7. Tactics, Techniques, and Procedures
8. Infrastructure and Indicators
9. Detection Engineering
 - 9.1. Detection Summary
 - 9.2. Sigma / Detection Content
 - 9.3. Hunt Query
10. Threat Hunting
 - 10.1. Hunt Hypothesis
 - 10.2. Hunt Query
11. Risk Assessment
12. Recommended Actions
13. References

Commercial Satellite and LEO Constellation Cyber Threat Landscape

1. Executive Summary

Public government and industry reporting warns that growing LEO satellite constellations and commercial satellite lifecycle dependencies expand attack surface across ground stations, mission operations, and downstream aviation/aerospace communications.

Lost Boys Cyber selected this item for the 2026-09-09 daily run because it affects practical defender priorities: satellite cyber, LEO, ground station, mission data. The report uses public-source reporting and translates it into detection, hunting, and executive action guidance without asserting private victim details or unpublished IoCs.

2. Intelligence Requirements

Question	Current Answer	Confidence
What activity is most important?	satellite cyber, LEO, ground station, mission data.	Medium
Who should care today?	Security teams responsible for exposed enterprise services, identity controls, endpoint telemetry, and sector-specific operational resilience.	Medium-High
What should defenders do first?	Validate exposure, harden identity/edge controls, and run behavior-focused hunts below.	High

3. Source Review & Confidence

Source	Contribution	Confidence
Breaking Defense	Primary selected article/report for this daily item.	Medium-High
CISA / ATT&CK / vendor context	Defensive control and technique mapping context.	High
Industry trend reports	Sector and operational context.	Medium

The selected public sources provide adequate depth for defensive prioritization. Attribution and exact IoCs are intentionally limited unless present in the cited reporting.

4. Threat Activity Overview

The activity is best understood as a defender-facing behavior cluster rather than a guaranteed single intrusion set. Key observed or reported characteristics include: satellite cyber, LEO, ground station, mission data. Operators should focus on behaviors visible in endpoint, identity, network, SaaS, and web telemetry.

5. Affected Sectors and Exposure

Exposure Area	Why It Matters
Internet-facing applications and network edge	Frequently abused for initial access and persistence.
Identity/help-desk workflows	Social engineering and MFA fatigue can bypass otherwise strong perimeter controls.

Endpoint and browser sessions	Browser injection, remote tooling, and script execution can evade perimeter-only controls.
Third-party dependencies	Shared vendors and SaaS workflows create cross-organization blast radius.

6. Aviation / Aerospace Sector Impact Analysis

Sector Segment	Operational Relevance	Likely Exposure Path	Third-Party / Supply-Chain Consideration	Defender Action
Airlines / airports	Disruption can affect check-in, baggage, crew scheduling, passenger trust, and regulatory reporting.	Identity compromise, exposed remote access, managed service provider compromise, vulnerable edge services.	Shared passenger-processing and managed IT vendors can propagate impact across multiple operators.	Segment critical airport/airline workflows, validate recovery runbooks, and test manual fallback.
Aerospace / defense manufacturing	Theft of engineering data or supplier access can support espionage and long-cycle intrusion.	Phishing, supplier portals, VPN appliances, router/firewall compromise, developer tooling.	Smaller suppliers may have weaker monitoring while holding sensitive program data.	Require MFA, supplier access reviews, export-control-aware logging, and edge-device patch SLAs.
Satellite / space systems	Ground-station and mission-network compromise can degrade communications and downstream services.	Ground segment IT, remote access, software supply chain, credential theft, and exposed APIs.	Commercial satellite services can be dependencies for aviation communications and tracking.	Separate enterprise IT from mission networks; monitor privileged sessions and command channels.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, social engineering, exposed service abuse, or supply-chain exposure	Consistent with selected public reporting and sector exposure model.
Execution	Script, browser, or remote-management tool execution	Relevant to reported behavior: satellite cyber, LEO, ground station, mission data.
Command and Control	Legitimate-service abuse or rare outbound connectivity	Public-cloud/service abuse and backdoor/RAT patterns require egress monitoring.
Impact / Collection	Credential theft, data theft, ransomware, or operational disruption	Depends on item; prioritize crown-jewel and operational workflows.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
No atomic IoCs asserted in this automated report	Source limitation	Use behavior-based analytics and check cited primary source for updated IoCs.
Legitimate cloud/SaaS services used unusually	Behavioral observable	Hunt for rare service use by unusual processes/accounts.
Remote access tooling in user sessions	Tooling pattern	Identify unauthorized help-desk or attacker-

controlled sessions.

9. Detection Engineering

This detection content is intentionally behavior-focused so it remains useful when public reporting omits hashes or domains.

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Script interpreter launched by browser, chat, office, or remote-support parent	EDR process telemetry	Catch social-engineering-to-execution chains.
Rare outbound connection by scripting or remote-management tool	Network/EDR telemetry	Surface C2 or payload staging.
New OAuth/session or admin action after help-desk contact	Identity/SaaS audit logs	Detect identity-led intrusion.

9.2. Sigma / Detection Content

```

title: Suspicious User-Driven Script Or Remote Tool Execution
status: experimental
description: Detects execution chains consistent with phishing, vishing, browser injection, or operator-assisted intrusion.
logsource:
  product: windows
  category: process_creation
detection:
  selection_parent:
    ParentImage|endswith:
      - '\chrome.exe'
      - '\msedge.exe'
      - '\firefox.exe'
      - '\Teams.exe'
      - '\OUTLOOK.EXE'
  selection_child:
    Image|endswith:
      - '\powershell.exe'
      - '\pwsh.exe'
      - '\cmd.exe'
      - '\wscript.exe'
      - '\mshta.exe'
      - '\AnyDesk.exe'
      - '\TeamViewer.exe'
  condition: selection_parent and selection_child
level: high

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "firefox.exe", "Teams.exe", "OUTLOOK.EXE")
| where FileName in~ ("powershell.exe", "pwsh.exe", "cmd.exe", "wscript.exe", "mshta.exe", "AnyDesk.exe", "TeamViewer.exe")
| project Timestamp, DeviceName, AccountName, InitiatingProcessFileName, FileName, ProcessCommandLine

```

10. Threat Hunting

10.1. Hunt Hypothesis

If this threat pattern touched the environment, telemetry should show abnormal user-assisted execution, remote-management installation, suspicious SaaS/session events, rare outbound traffic, or atypical administrative changes close to the article publication window.

10.2. Hunt Query

```
DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("powershell.exe", "pwsh.exe", "cmd.exe", "wscript.exe",
"mshta.exe", "AnyDesk.exe", "TeamViewer.exe", "chrome.exe", "msedge.exe")
| summarize Connections=count(), RemoteIPs=make_set(RemoteIP, 25), RemoteUrls=make_set(RemoteUrl, 25)
by DeviceName, InitiatingProcessFileName, AccountName, bin(Timestamp, 1h)
| where Connections > 10 or array_length(RemoteIPs) > 5

SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType == 0
| summarize Countries=make_set(Location, 20), Apps=make_set(AppDisplayName, 20),
IPs=make_set(IPAddress, 20), Signins=count() by UserPrincipalName, bin(TimeGenerated, 1d)
| where array_length(Countries) > 2 or Signins > 50
```

11. Risk Assessment

Dimension	Assessment
Likelihood	Medium to High where exposed services, weak MFA, unmanaged remote tools, or high-value sector operations exist.
Impact	Ranges from credential theft and data loss to operational disruption depending on environment and threat variant.
Priority	Review today if the primary telemetry or affected sector matches your environment.

12. Recommended Actions

Priority	Owner	Action
High	SOC	Run the KQL hunts and review any matching execution chains.
High	IAM	Review MFA, help-desk verification, session anomalies, and admin consent activity.
Medium	Network	Baseline remote-support tools, cloud-service egress, and critical third-party connectivity.
Medium	IR / BC	Validate containment and manual fallback procedures for operationally critical workflows.

13. References

- [1] Breaking Defense: <https://breakingdefense.com/2026/03/new-joint-intel-report-warns-of-cyber-threats-to-growing-leo-satellite-constellations/>

- [2] ENISA Space Threat Landscape: <https://www.enisa.europa.eu/news/from-cyber-to-outer-space-a-guide-to-securing-commercial-satellite-operations>
- [3] Reuters ENISA airport ransomware confirmation: <https://www.reuters.com/business/aerospace-defense/eu-agency-says-third-party-ransomware-behind-airport-disruptions-2025-09-22/>
- [4] CybelAngel aviation cyberattacks impact: <https://cybelangel.com/blog/the-global-impact-of-aviation-cyberattacks/>